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Systems and methods for on-network device identification

Abstract

Systems and methods are described for on-network device identification. A user device may make an application programming interface (API) call to an address associated with a network device. The API call may comprise an indication of an address associated with the user device. Based on the API call, the network device may send an indication of the address to a computing device associated with an on-network service. The computing device associated with the on-network service may determine that the user device is authorized to connect to a network or to access an on-network service.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/647,609, filed Jan. 11, 2022, now U.S. Pat. No. 11,750,613, which is a continuation of U.S. patent application Ser. No. 16/046,804, filed Jul. 26, 2018, now U.S. Pat. No. 11,252,155, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) A service provider may comprise computing devices, such as servers, in a network for providing on-network services. A subscriber of the service provider may use a user device to connect with the network via a network device at a premises. The user device may have an address. A network device may use a mask preventing a service provider computing device from determining the user device address. It may be beneficial for the service provider computing device to obtain the user device address.

SUMMARY

(2) Systems and methods are described for on-network device identification. An example method may comprise causing a user device to make an application programming interface (API) call to an address associated with a network device. The API call may comprise an indication of an address associated with the user device. The example method may comprise receiving, based on the API call, an indication that the user device is configured to communicate with the network device via a local area network.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following drawings show generally, by way of example, but not by way of limitation, various examples discussed in the present disclosure. In the drawings:

- (2) FIG. 1 is a block diagram of an example premises.
- (3) FIG. 2 is an example sequence diagram.
- (4) FIG. 3 is an example sequence diagram.
- (5) FIG. 4 is an example sequence diagram.
- (6) FIG. 5 is a flow diagram of an example method.
- (7) FIG. 6 is a flow diagram of an example method.
- (8) FIG. 7 is a flow diagram of an example method.
- (9) FIG. 8 is a flow diagram of an example method.
- (10) FIG. 9 is a block diagram of an example system environment.

DETAILED DESCRIPTION

(11) Systems and methods are described for on-network device identification. A network device may be located at a premises. The network device may comprise any device (e.g., a gateway device, a cable modem, a wireless router, etc.) for connecting user devices (e.g., smart phones, tablets, laptops, computing devices, etc.) within the premises with a network, such as the Internet. The network device may comprise instructions (e.g., a program, software, firmware, etc.) for determining an address (e.g., media access control (MAC) address, local internet protocol (IP) address, etc.) for each user device attached to (e.g., connected to, in communication with, etc.) the

network device.

(12) Access to the network via the network device may be provided by a computing device associated with a service provider. The network may comprise service provider computing devices, such as servers. The computing devices may provide on-network services to the network device and/or the user devices located at the premises and attached to the network device. A web application programming interface (API) may provide commands (e.g., calls, actions, etc.) to facilitate communication between the computing devices and the network device and/or the user devices located at the premises and attached to the network device. The web API may not be able to determine an address (e.g., MAC address, local IP address) for a user device attached to the network device. A user device may make an API call to an IP address associated with the network device. The API call may comprise an indication of an address (e.g., MAC address, local IP address) associated with the user device. Based on the API call, the network device may generate a message with an indication that the user device associated with the address in the API call is attached to the network device. The network device may send the generated message to the user device and/or to the computing device.

(13) FIG. 1 is a block diagram **100** of an example premises **102**. The premises **102** may comprise a user device **104** and a network device **106**. Although only one user device **104** is shown, the premises **102** may comprise more than one user device **104**. Although only one network device **106** is shown, the premises **102** may comprise more than one network device **106**.

(14) The user device **104** may comprise a smart phone, a tablet, a laptop, a wearable computing device, or any other appropriate computing device. The user device **104** may be associated with an account (such as a service provider account). The account may be or may not be associated with the premises **102**. The account may be associated with a subscription to on-network services. The on-network services may comprise digital content services, such as services providing videos, music, books, images, or text. The on-network services may comprise network connectivity services, such as services that configure the user device **104** to connect to one or more networks. The on-network services may comprise applications, such as mobile applications or other user applications. The on-network services may comprise internet of things services, home automation services, security system services, or premises management services. The on-network services may comprise interfaces, permissions, or configuration of the user-device enabling control of other devices via the user device **104**. The on-network services may be associated with or provided by a computing device **110**, such as via a network **108**. The computing device **110** may be associated with a service provider. The computing device **110** may be associated with one or more on-network services. The computing device **110** may be external to the premises **102**. The user device **104** may not be associated with an account or a subscription to on-network services.

(15) The user device **104** may have an associated address (e.g., MAC address, local IP address, etc.). The user device **104** may be in communication with the network device **106**. The user device **104** and the network device **106** may communicate using Institute of Electrical and Electronics Engineers (IEEE) 802.11, Bluetooth, or any other communication protocol appropriate for communication between two devices local to the premises **102**.

(16) The network device **106** may comprise a cable modem, a wireless router, a gateway device, or any device that provides the premises **102** access to the network **108**, such as the Internet. The network device **106** may communicate with other devices, such as a computing device **110** via the network **108**. The network device **106** may communicate with the computing device **110** using internet protocol (IP) or any other communication protocol appropriate for communication between two devices communicating over the network **108**. The network device **106** may facilitate communication between the user device **104** and the computing device **110**. One or more application programming interfaces (APIs) may be used in communications between the user device **104**, the network device **106**, and the computing device **110**.

(17) The computing device **110** may provide the one or more on-network services for one or more

user devices **104** attached to the network device **106** at the premises **102**. The computing device **110** may use the address associated with the user device **104** to determine that the user device **104** is authorized to access or receive the one or more on-network services.

(18) The computing device **110** may send a prompt message through the network **108** to the network device **106**. The network device **106** may forward the prompt message to the user device **104** attached to the network device **106**. The prompt message may cause the user device **104** to make an API call to the network device **106**. The API call may comprise an indication of the address associated with the user device **104**. Based on the API call, the network device **106** may generate an indication that the user device **104** is attached to the network device **106**. The indication may comprise an indication of the address associated with the user device **104**. The network device **106** may forward the indication of the address to the computing device **110** and/or to the user device **104**. The user device **104** may forward the indication of the address to the computing device **110** via the network device **106**.

(19) Based on the indication of the address associated with the user device **104**, the computing device **110** may determine that the user device **104** is attached to the network device **106**. Based on the indication of the address associated with the user device **104**, the computing device **110** may determine that the user device **104** is associated with a subscriber of the on-network services. Based on the indication of the address associated with the user device **104** and/or the indication that the user device **104** is attached to the network device **106**, the computing device **110** may determine that the user device **104** is located at the premises **102** associated with a subscriber of the on-network services. Based on the determination that the user device **104** is attached to the network device **106**, the determination that the user device is associated with a subscriber, and/or the determination that the user device is located at the premises **102**, the computing device **110** may determine that the user device **104** is authorized to access one or more of the on-network services.

(20) It may be determined that the user device **104** is authorized to access one or more on-network services based on a profile associated with the user device **104**. The profile may indicate one or more on-network services or one or more types of on-network services that the user device **104** is authorized to access. The profile may be determined by a user, such as of the user device **104** or the network device **104**. For example, a parent may configure the profile to exclude on-network services that are not intended for children. The profile may be determined by the service provider. For example, the user device **104** may be authorized to access content that is included with a subscription package. The user device **104** may not be authorized to access premium content that is outside the subscription package, unless the user upgrades to a more expansive subscription package. The profile may be determined based on the user device **104** making the API call to the network device, such as by the network device **106** or the network **108**.

(21) The computing device **110**, may provide the user device **104** with the on-network service, such as via the network **108**. The computing device **110** may provide the user device **104** with the on-network service via the network device **106**. For example, the computing device **110** may stream a video to the network device **106** and the network device **106** may stream the video to the user device **104**.

(22) The computing device **110** may determine to grant a different user device located at a premises associated with the user device **104** access to other devices located at the premises associated with the user device **104**. The computing device **110** may determine to grant a different user device located at a premises associated with the user device **104** access to other devices located at the premises associated with the user device **104** based on the determination that the user device **104** is located at the premises **102**. The other devices may comprise home automation devices, security system devices, internet of things devices, and/or premises management devices. For example, based on the determination that the user device is located at the premises **102**, the computing device **110** may determine that the user device **104** is not located at the home of a user of the user device **104**. The user may be traveling and a caretaker may be at the home of the user. The

computing device **110** may determine to enable a user device of the caretaker to control home automation devices at the home of the user.

(23) FIG. 2 shows an example sequence diagram **200**. At **202**, the computing device **110** of FIG. 1 may send a prompt message to the network device **106** of FIG. 1. The prompt message may comprise an indication that the user device **104** requests communication with the network device **106**. The prompt message may comprise an indication of an address of the user device **104**. The prompt message may comprise an indication that the user device **104** made an API call to the network device **106**.

(24) At **204**, the network device **106** may forward the prompt message to the user device **104** of FIG. 1. Based on the prompt message, the user device **104** may make an API call to the network device **106** at **206**. The API call may comprise an indication of an address (e.g., MAC address, local IP address, etc.) associated with the user device **104**. Based on the API call, the network device **106** may generate an indication that the user device **104** is attached to the network device **106**. The indication may comprise the address associated with the user device **104**. The indication may comprise a profile associated with the user device **104**. The profile may indicate one or more on-network services or one or more types of on-network services that the user device **104** is authorized to access. The profile may be determined by a user, such as of the user device **104** or the network device **104**. The profile may be determined by the service provider. At **208**, the network device **106** may send the indication to the computing device **110**.

(25) A content distribution server (e.g., computing device **110**) may send, to a gateway device (e.g., network device **106**) located at premises (e.g., premises **102**), a first prompt message comprising a request for an indication of user devices in communication with the gateway device. Based on the first prompt message, the gateway device may send a second prompt message to a laptop computer (e.g., user device **104**) in communication with the gateway device and located at the premises. Based on the second prompt message, the laptop may make an API call to the gateway device comprising an indication of an IP address of the laptop. Based on the API call, the gateway device may send an indication of the IP address of the laptop to the content distribution server.

(26) Based on the indication of the IP address of the laptop, the content distribution server may determine that the laptop is located at the premises with the gateway device and in communication with the gateway device. Based on the determination that the laptop is located at the premises with the gateway device and in communication with the gateway device, the content distribution server may determine that the laptop is authorized to access a content delivery service (e.g., on-network services). Based on the determination that the laptop is authorized to access the content delivery service, the content distribution server may stream content associated with the content delivery service to the laptop.

(27) FIG. 3 is an example sequence diagram **300**. At **302**, the user device **104** of FIG. 1 may make an API call to the network device **106** of FIG. 1. The user device **104** may make the API call in connecting to the network device **106** or in attempting to access on-network services provided by the computing device **110**. The API call may comprise an indication of an address (e.g., MAC address, local IP address, etc.) associated with the user device **104**.

(28) The network device **106** may generate an indication that the user device **104** is attached to the network device **106**, such as based on the API call. The indication may comprise the address associated with the user device **104**. At **304**, the network device **106** may send the indication to the user device **104**. At **306**, the user device **104** may make another API call to the network device. The user device **104** may make the API call in requesting access to one or more of the on-network services. At **308**, the network device **106** may communicate with the computing device **110**, such as based on the other API call. The network device **106** may send an indication of the address associated with the user device **104** to the computing device **110**. The network device **106** may send an indication of the request by the user device **104** for the on-network service. The computing device **110** may determine that the user device **104** is authorized to access the on-network service,

such as based on the indication sent in **308**. The computing device **110** may provide the on-network service.

(29) Additionally or alternatively, a first user may bring a tablet device (e.g., user device **104**) to a second user's house (e.g., premises **102**). The first user may an input, via the tablet, a selection to connect the tablet to a router (e.g., network device **106**) or to a network of the router. The tablet may make an API call to the router, such as based on the selection. The API call may comprise an indication of a MAC address of the tablet. The router may send an indication of the MAC address and/or an indication that the tablet is connected to the router to the computing device **110**, such as based on the API call.

(30) FIG. **4** is an example sequence diagram **400**. At **402**, the user device **104** of FIG. **1** may make an API call to the network device **106** of FIG. **1**. The API call may comprise a request to connect to a network at the premises **102**. The API call may comprise an indication of an address (e.g., MAC address, local IP address, etc.) associated with the user device **104**.

(31) At **403**, the network device **106** may register the user device **104** with a service provider. The network device **106** may register the user device **104** with a service provider based on the API call. The service provider may be associated with the network, such as a provider of the network or a provider of on-network services via the network. The user device **104** may or may not be associated with a subscriber of the service provider. Registering the user device **104** may comprise sending an indication of the address of the user device **104** to the computing device **110**.

Registering the user device **104** may comprise sending, to the service provider server, an indication that the user device **104** is connected to the network device **106**.

(32) The computing device **110** may determine that the user device **104** is associated with a subscriber of the service provider, such as based on the indication of the address. The computing device **110** may determine that the user device **104** is authorized to connect to the network at the premises **102**, such as based on the determination that the user device is associated with the subscriber.

(33) Based on the indication that the user device is connected to the network device **106**, the service provider server may determine that the user device **104** is located at the premises **102** where the network device is located. Based on the determination that the user device **104** is located at the premises, the computing device **110** may determine that the user device **104** is authorized to connect to the network at the premises **102**. The computing device **110** may store an indication of the user device **104**, such as the address of the user device **104**, in a registry of authorized devices.

(34) At **404**, the network device **106** may send, to the user device **104**, an indication that the user device is authorized to connect to the network. The network device **106** may send the user device **104** and indication that the user device **104** is registered as an authorized device with the service provider. The network device **106** may send the user device **104** and indication of a profile associated with the user device **104**. The profile may be associated with filters, permission, or rules associated with on-network services that the user device **104** is authorized to access. The profile may be determined by the service provider, the computing device **110**, the network device **106**, or a user of the user device **104**.

(35) At **405**, the user device **104** may send, to the network device **106**, a request to access one or more of the on-network services. The request may comprise an indication of the address of the user device **104**. At **406**, the network device **106** may send an indication of the request to the computing device **110**, such as based on the request. The indication of the request may comprise an indication of the address associated with the user device **104**. The indication of the request may comprise an indication of the requested on-network services. The computing device **110** may determine that the user device **104** is authorized to access the requested on-network service, such as based on the indication of the request. Authorizing the user device **104** may comprise determining that the registry of authorized devices comprises the indication of the user device, such as the address associated with the user device. Authorizing the user device **104** may be based on the profile

associated with the user device **104**.

(36) At **407**, the computing device **110** may provide the on-network service to the user device **104**. The computing device **110** may provide the on-network service to the user device **104** based on the determination that the user device **104** is authorized to access the requested on-network service. Providing the requested on-network service may comprise storing data or content associated with the requested on-network service at a storage location available to the user device **104**. Providing the requested on-network service may comprise sending data or content associated with the requested on-network service to the user device **104** directly or via another device. For example, the service provider server **110** may send the data or content associated with the requested on-network service to the network device **106**. At step **408**, the network device **106** may send the data or content associated with the requested on-network service to the user device **104**.

(37) FIG. 5 shows an example method **500**. At step **510**, a user device **104** may be caused to make an application programming interface (API) call to an address associated with a network device **106**. The computing device **110** of FIG. 1 may cause the user device **104** of FIG. 1 to make an API call to an address associated with the network device **106** of FIG. 1. A user may cause the user device **104** to make the API call, such as by inputting a selection to connect to a network at the premises **102** or by inputting a selection to access one or more on-network services. The address associated with the network device **106** may comprise an Internet Protocol (IP) address. The network device **106** may comprise a gateway device or a wireless router. The API call may comprise an indication of an address associated with the user device **104**. The address associated with the user device **104** may comprise an Internet Protocol (IP) address. The address associated with the user device **104** may comprise a media access control (MAC) address.

(38) At step **520**, an indication that the user device **104** is configured to communicate with the network device **106** via a local area network may be received based on the API call. The computing device **110** may receive, based on the API call, an indication that the user device **104** is configured to communicate with the network device **106** via a local area network at the premises **102**. The indication that the user device **104** is configured to communicate with the network device **106** via the local area network may be generated based on the address associated with the user device **104**. The network device **106** may generate the indication that the user device **104** is configured to communicate with the network device **106** via the local area network based on the address associated with the user device **104**.

(39) At step **530**, the user device **104** may be granted access to the network at the premises **102**. The user device **104** may be granted access to an on-network service. The user device **104** may be granted access to the network at the premises **102** and/or to the on-network service based on the indication that the user device **104** is configured to communicate with the network device **106** via the local area network. The computing device **110** may grant, based on the indication that the user device **104** is configured to communicate with the network device **106** via a local area network, the user device **104** access to an on-network service. The on-network service may comprise a digital content service, a network connectivity service, an application, an internet of things service, a home automation service, a security system service, or a premises management service.

(40) Additionally or alternatively, a user may be a subscriber to a service provider. The user may have a mobile phone (e.g., user device **104**). The mobile phone may have an IP address. The service provider may associate the IP address with the mobile phone and/or the subscriber/user. The user may go to a home (e.g., premises **102**) of another subscriber of the service provider. The home may comprise a gateway device (e.g., network device **106**) configured to connect to a network (e.g., network **108**). The mobile phone may connect to the gateway device to access the network. The network may be provided by a computing device associated with a service provider and/or an on-network service (e.g., computing device **110**). The computing device may send a prompt to the gateway device. In response, the gateway device may send the prompt to the mobile phone. The prompt may cause the mobile phone to make an API call to an address associated with the gateway

device. The API call may comprise an indication of the address associated with the mobile phone. Based on the API call, the gateway device may generate an indication that the mobile phone is attached to the gateway device. The indication may comprise the address associated with the mobile phone. The gateway device may send the indication that the mobile phone is attached to the gateway device to the computing device. Based on the indication that the mobile phone is attached to the gateway device and/or the indication of the address associated with the mobile phone, the computing device may determine that the mobile phone associated with the subscriber is at the house associated with the other subscriber. Based on the determination that the mobile phone associated with the subscriber is at the house associated with the other subscriber, the computing device may provide the mobile phone with an on-network service.

(41) FIG. 6 shows an example method 600. At step 610, an application programming interface (API) call may be received by a network device 106, from a user device 104, at an address associated with the network device 106. The address associated with the network device 106 may comprise an Internet Protocol (IP) address. The network device 106 of FIG. 1 may receive, from the user device 104 of FIG. 1, an API call at an address associated with the network device 106. The API call may comprise an indication of an address associated with the user device. The network device 106 may comprise a gateway device or a wireless router. The address associated with the user device 104 may comprise an Internet Protocol (IP) address. The address associated with the user device 104 may comprise a media access control (MAC) address.

(42) At step 620, an indication that the user device 104 is configured to communicate with the network device 106 may be generated. An indication that the user device 104 is configured to communicate with the network device 106 via a local area network may be generated. The indication may be generated based on the address. For example, the indication may be generated based on a determination that the address is associated with a user and/or an account authorized to communicate with the network device 106, communicate via the local area network, and/or access an on-network device. The indication may be generated based on the API call. The indication may be generated by the network device 106.

(43) At step 620, the indication that the user device 104 is configured to communicate with the network device 106 via the local area network may be sent based on the API call. The network device 106 may send, based on the API call, an indication that the user device 104 is configured to communicate with the network device 106 via a local area network to the user device 104 and/or the computing device 110 of FIG. 1. The indication that the user device 104 is configured to communicate with the network device 106 via the local area network may be generated based on the address associated with the user device 104. The network device 106 may generate the indication that the user device 104 is configured to communicate with the network device 106 via the local area network based on the address associated with the user device 104.

(44) For example, a subscriber to a service provider may reside at a premises 102. The premises 102 may comprise a network device 106 configured to connect to a network. The network may comprise a computing device associated with a service provider and/or an on-network service. Another subscriber of the service provider may come to the premises 102. The other subscriber may comprise a user device 104. The user device 104 may have an address. The service provider may associate the address with the user device 104 and/or the other subscriber. The user device 104 may connect to the network device 106 to access the network. The network device 106 may receive an API call from the user device 104. The API call may comprise an indication of the address associated with the user device 104. The network device 106 may generate an indication that the user device 104 is attached to the network device 106. The indication may comprise the address associated with the user device 104. The network device 106 may send the indication to the computing device 110 and/or the user device 104. Based on the indication that the user device 104 is attached to the network device 106 and/or the indication of the address associated with the user device 104, the computing device 110 may determine that the user device 104 associated with the

subscriber is at the premises **102** associated with the other subscriber. Based on the determination that the user device **104** associated with the subscriber is at the premises **102** associated with the other subscriber, the computing device **110** may provide the user device **104** with an on-network service.

(45) FIG. 7 shows an example method **700**. At step **710**, a user device **104** may make an application programming interface (API) call to an address associated with a network device **106**. The address associated with the network device **106** may comprise an Internet Protocol (IP) address. The user device **104** of FIG. 1 may send an API call to an address associated with the network device **106** of FIG. 1. The user device **104** may make the API call to connect to a network at a premises **102**. The API call may comprise an indication of an address associated with the user device **104**. The network device **106** may comprise a gateway device or a wireless router. The address associated with the user device **104** may comprise an Internet Protocol (IP) address. The address associated with the user device **104** may comprise a media access control (MAC) address.

(46) At step **720**, an indication that the user device **104** is configured to communicate with the network device **106** via a local area network may be received based on the API call. The user device **104** may receive, based on the API call, an indication that the user device **104** is configured to communicate with the network device **106** via a local area network. The indication that the user device **104** is configured to communicate with the network device **106** via the local area network may be generated based on the address associated with the user device **104**. The network device **106** may generate the indication that the user device **104** is configured to communicate with the network device **106** via the local area network based on the address associated with the user device **104**.

(47) An indication of a profile associated with the user device **104** may be received. The profile may indicate one or more on-network services or one or more types of on-network services that the user device **104** is authorized to access. The profile may be determined by a user, such as of the user device **104** or the network device **104**. For example, a parent may configure the profile to exclude on-network services that are not intended for children. The profile may be determined by the service provider. The user device **104** may be authorized to access content that is included with a subscription package. The user device **104** may not be authorized to access premium content that is outside the subscription package, unless the user upgrades to a more expansive subscription package. The profile may be determined based on the user device **104** making the API call to the network device, such as by the network device **106** or the network **108**.

(48) At step **730**, an indication may be sent to a service provider to enable access to an on-network service. The user device **104** may send a request to the computing device **110** of FIG. 1 to enable access to an on-network service. The on-network service may a digital content service, a network connectivity service, an application, an internet of things service, a home automation service, a security system service, or a premises management service.

(49) For example, a user may be a subscriber of a service provider. The user may have a user device **104**. The user device **104** may have an address. The service provider may associate the address with the user device **104** and/or the subscriber. The user may go to a premises **102** of another subscriber of the service provider. The premises **102** may comprise a network device **106** configured to connect to a network. The user device **104** may connect to the network device **106** to access the network. The network may comprise a computing device **110**. The user device **104** may make an API call to the network device **106**. The API call may comprise an indication of the address. The network device **106** may generate an indication that the user device **104** is attached to the network device **106**. The indication may comprise the address associated with the user device **104**. The network device **106** may send the indication to the user device **104**. The user device **104** may send the indication to the computing device **110** via the network device **106**. Based on the indication that the user device **104** is attached to the network device **106** and/or the indication of the address associated with the user device **104**, the computing device **110** may determine that the

user device **104** associated with the subscriber is at the premises **102** associated with the other subscriber. Based on the determination that the user device **104** associated with the subscriber is at the premises **102** associated with the other subscriber, the computing device **110** may provide the user device **104** with an on-network service, such as a content service.

(50) For example, a user of a user device **104** may not be a subscriber of a service provider. Based on a determination that the user device **104** is located at a premises **102** associated with a subscriber or is in communication with a network device **106** associated with a subscriber, the computing device **110** may determine to authorize the user device **104** to connect to a network or to access on-network services. The computing device **110** may determine to authorize the user device **104** to connect to the network or access the on-network services while the user device **104** is at the premises **102** or connected to the network device **106**. The computing device **110** may determine that the user device **104** remains at the premises by determining that the user device **104** remains in communication with the network device **106** or by providing the on-network services to the user device **104** via the network device **106**.

(51) FIG. **8** shows an example method **800**. At step **810**, a request to connect to a network at a premises **102** may be received from a user device **104**. The request may comprise an API call, such as to a network device **106**. The user device **104** and the network device **106** may be located at the premises **102**. The network may comprise a local area network. The request may comprise an address (e.g., MAC address, IP address) associated with the user device **104**. The user device may be associated with a subscriber of a service provider. The user device may not be associated with a subscriber of a service provider. For example, the user of a user device may not be a subscriber of a service provider, but may be visiting the house or business of a subscriber of the service provider.

(52) At step **820**, based on the request to connect to the network, the network device **106** may send an indication of the request to a computing device **110**. The indication may comprise an indication of the address associated with the user device **104**. Based on the indication, the computing device **110** may determine that the user device **104** is located at the premises and/or that the user device is in communication with the network device **106**. Based on the determination that the user device **104** is located at the premises and/or that the user device is in communication with the network device **106**, the computing device **110** may store an indication of the user device **104** in a registry. The registry may be associated with devices authorized to connect to the network and/or devices authorized to access on-network services. The indication of the user device **104** may comprise an indication of the address associated with the user device **104** and/or an indication of a subscriber account associated with the user device **104**.

(53) Based on the address associated with the user device **104**, the computing device **110** may determine that the user device **104** is associated with a subscriber of a service provider associated with the computing device **110**. Based on the determination that the user device **104** is associated with the subscriber of a service provider associated with the computing device **110**, the computing device **110** may store the indication of the user device **104** on the registry. Based on one or more of the determination that the user device **104** is located at the premises, the determination that the user device is in communication with the network device **106**, and/or the determination that that the user device **104** is associated with a subscriber of a service provider associated with the computing device **110**, the computing device **110** may enable the user device **104** to connect to the network. Enabling the user device **104** to connect to the network may comprise sending an indication to the network device **106** to connect to the network or to enable the user device **104** to connect to the network. An indication that the user device **104** is connected to the network and/or to the network device **106** may be send to the user device **104**, such as by the network device **106**.

(54) A profile associated with the user device **104** may be determined. The profile may indicate one or more on-network services or one or more types of on-network services that the user device **104** is authorized to access. The profile may be determined by a user, such as of the user device **104** or the network device **104**. For example, a parent may configure the profile to exclude on-network

services that are not intended for children. The profile may be determined by the service provider. The user device **104** may be authorized to access content that is included with a subscription package. The user device **104** may not be authorized to access premium content that is outside the subscription package, unless the user upgrades to a more expansive subscription package. An indication of the profile may be stored, such as to the registry. An indication of the profile may be sent to the user device **104**, such as by the network device **106** or the computing device **110**.

(55) At step **830**, a request may be received, from the user device, to access one or more on-network services. The on-network services may comprise digital content services, network connectivity services, mobile applications, internet of things services, home automation services, security system services, or premises management services. The request may comprise an indication of the address associated with the user device **104**. The request may comprise an API call to the network device **106**.

(56) At step **840**, based on the request to access the on-network services, the network device **106** may send an indication of the request to the computing device **110**. The indication of the request may comprise an indication of the services. The indication of the request may comprise an indication of the address associated with the user device **104**.

(57) Based on the indication of the request, the computing device **110** may determine whether the user device **104** is authorized to access the on-network services. The computing device **110** may determine that the registry comprises the indication of the user device **104**. Based on the determination that the registry comprises the indication of the user device **104**, the computing device **110** may determine that the user device is authorized to access the on-network services. The computing device **110** may determine whether the user device **104** is authorize to access the requested on-network services based at least on the profile associated with the user device **104**. The computing device **110** may determine whether the user device **104** is authorize to access the requested on-network services based at least on a subscription of a user of the user device **104**.

(58) At step **850**, based on the determination that the user device **104** is authorized to access the requested on-network services, the requested on-network services may be provided to the user device **104**, such as by the computing device **110**. Providing the requested on-network service may comprise storing data or content associated with the requested on-network service at a storage location available to the user device **104**. Providing the requested on-network service may comprise sending data or content associated with the requested on-network service to the user device **104** directly or via another device. For example, the computing device **110** may send the data or content associated with the requested on-network service to the network device **106**. At step **408**, the network device **106** may send the data or content associated with the requested on-network service to the user device **104**.

(59) FIG. **9** shows an example operating environment **900**. The example operating environment **900** is not intended to suggest any limitation as to the scope of use or functionality of operating environment architecture. Neither should the operating environment be interpreted as having any dependency or requirement relating to any one or combination of components shown in the example operating environment.

(60) The present methods and systems may be operational with numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with the systems and methods comprise, but are not limited to, personal computers, server computers, laptop devices, and multiprocessor systems. Additional examples comprise set top boxes, programmable consumer electronics, network PCs, minicomputers, and mainframe computers, distributed computing environments that comprise any of the above systems or devices, and the like.

(61) The processing of the disclosed methods and systems may be performed by software components. The disclosed systems and methods may be described in the general context of

computer-executable instructions, such as program modules, being executed by one or more computers or other devices. Generally, program modules comprise computer code, routines, programs, objects, components, data structures, etc. that performs particular tasks or implements particular abstract data types. The disclosed methods may be practiced in grid-based and distributed computing environments where tasks may be performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules may be located in both local and remote computer storage media including memory storage devices.

(62) Additionally or alternatively, the systems and methods disclosed may be implemented via a general-purpose computing device in the form of a computing device **901**. The components of the computing device **901** may comprise, but are not limited to, one or more processors or processing units **903**, a system memory **912**, and a system bus **913** that couples various system components including the processor **903** to the system memory **912**. In the case of multiple processing units **903**, the system may utilize parallel computing.

(63) The system bus **913** represents one or more of several possible types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, such architectures may comprise an Industry Standard Architecture (ISA) bus, a Micro Channel Architecture (MCA) bus, an Enhanced ISA (EISA) bus, a Video Electronics Standards Association (VESA) local bus, an Accelerated Graphics Port (AGP) bus, and a Peripheral Component Interconnects (PCI), a PCI-Express bus, a Personal Computer Memory Card Industry Association (PCMCIA), Universal Serial Bus (USB) and the like. The bus **913**, and all buses specified in this description may be implemented over a wired or wireless network connection and each of the subsystems, including the processor **903**, a mass storage device **904**, an operating system **905**, on-network device identification software **906**, on-network device identification data **907**, a network adapter **908**, system memory **912**, an Input/output Interface **910**, a display adapter **909**, a display device **911**, and a human machine interface **902**, may be contained within one or more remote computing devices **914a,b,c** at physically separate locations, connected through buses of this form, in effect implementing a fully distributed system.

(64) The computing device **901** typically comprises a variety of computer readable media. Example readable media may comprise any available media that is accessible by the computing device **901** and may comprise both volatile and non-volatile media, removable and non-removable media. The system memory **912** comprises computer readable media in the form of volatile memory, such as random access memory (RAM), and/or non-volatile memory, such as read only memory (ROM). The system memory **912** typically contains data such as on-network device identification data **907** and/or program modules such as operating system **905** and on-network device identification software **906** that are immediately accessible to and/or are presently operated on by the processing unit **903**.

(65) The computing device **901** may comprise other removable/non-removable, volatile/non-volatile computer storage media. By way of example, FIG. **9** shows a mass storage device **904** which may provide non-volatile storage of computer code, computer readable instructions, data structures, program modules, and other data for the computing device **901**. For example and not limitation, a mass storage device **904** may be a hard disk, a removable magnetic disk, a removable optical disk, magnetic cassettes or other magnetic storage devices, flash memory cards, CD-ROM, digital versatile disks (DVD) or other optical storage, random access memories (RAM), read only memories (ROM), electrically erasable programmable read-only memory (EEPROM), and the like.

(66) Optionally, any number of program modules may be stored on the mass storage device **904**, including by way of example, an operating system **905** and on-network device identification software **906**. Each of the operating system **905** and on-network device identification software **906** (or some combination thereof) may comprise elements of the programming and the on-network

device identification software **906**. On-network device identification data **907** may be stored on the mass storage device **904**. On-network device identification data **907** may be stored in any of one or more databases known in the art. Examples of such databases comprise, DB2®, Microsoft® Access, Microsoft® SQL Server, Oracle®, MySQL, PostgreSQL, and the like. The databases may be centralized or distributed across multiple systems.

(67) The user may enter commands and information into the computing device **901** via an input device (not shown). Examples of such input devices may comprise, but are not limited to, a keyboard, pointing device (e.g., a “mouse”), a microphone, a joystick, a scanner, tactile input devices such as gloves, and other body coverings, and the like. These and other input devices may be connected to the processing unit **903** via a human machine interface **902** that is coupled to the system bus **913**, but may be connected by other interface and bus structures, such as a parallel port, game port, an IEEE 1394 Port (also known as a Firewire port), a serial port, or a universal serial bus (USB).

(68) A display device **911** may be connected to the system bus **913** via an interface, such as a display adapter **909**. It is contemplated that the computing device **901** may have more than one display adapter **909** and the computer **901** may have more than one display device **911**. For example, a display device may comprise a monitor, an LCD (Liquid Crystal Display), or a projector. In addition to the display device **911**, other output peripheral devices may comprise components such as speakers (not shown) and a printer (not shown) which may be connected to the computing device **901** via Input/Output Interface **910**. Any step and/or result of the methods may be output in any form to an output device. Such output may comprise any form of visual representation, including, but not limited to, textual, graphical, animation, audio, tactile, and the like. The display **911** and computing device **901** may comprise part of one device, or separate devices.

(69) The computing device **901** may operate in a networked environment using logical connections to one or more remote computing devices **914a,b,c**. By way of example, a remote computing device may comprise a personal computer, portable computer, a smart phone, a server, a router, a network computer, a peer device or other common network node. Logical connections between the computing device **901** and a remote computing device **914a,b,c** may be made via a network **915**, such as a local area network (LAN) and a general wide area network (WAN). Such network connections may be through a network adapter **908**. A network adapter **908** may be implemented in both wired and wireless environments. Such networking environments are conventional and commonplace in dwellings, offices, enterprise-wide computer networks, intranets, and the Internet.

(70) For example, application programs and other executable program components such as the operating system **905** are shown herein as discrete blocks, although such programs and components may reside at various times in different storage components of the computing device **901**, and may be executed by the data processor(s) of the computer. An implementation of on-network device identification software **906** may be stored on or sent across some form of computer readable media. Any of the disclosed methods may be performed by computer readable instructions embodied on computer readable media. Computer readable media may comprise any available media that may be accessed by a computer. By way of example and not limitation, computer readable media may comprise “computer storage media” and “communications media.” “Computer storage media” comprise volatile and non-volatile, removable and non-removable media implemented in any methods or technology for storage of information such as computer readable instructions, data structures, program modules, or other data. Example computer storage media may comprise RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which may be used to store the desired information and which may be accessed by a computer.

Claims

1. A method comprising: sending, by a computing device of a service provider that provides at least one service, to a network device located at a premises, a request configured to cause the network device to cause a device located at the premises to send, to the network device, an indication that the device is configured to communicate with the network device at the premises, wherein the premises is associated with a user account that is authorized to access the at least one service; receiving, from the network device, the indication; based on the indication, determining that the device is located at the premises; and based on determining that the device is located at the premises, granting the device access to the at least one service.
2. The method of claim 1, wherein sending the request comprises sending, to the network device, a first prompt message that causes the network device to send a second prompt message to the device, the first prompt message comprising at least in part the request for an indication of devices in communication with the network device.
3. The method of claim 1, wherein the indication comprises an address associated with the device.
4. The method of claim 1, wherein the indication indicates that the device is connected to the network device via a local area network.
5. The method of claim 1, further comprising sending, based on the granting the device access to the at least one service, to the network device, content associated with the at least one service to cause the network device to forward the content associated with the at least one service to the device.
6. The method of claim 1, further comprising storing, based on the granting the device access to the at least one service, in a memory associated with the computing device, data indicating that the device is authorized to access the at least one service.
7. The method of claim 6, wherein the indication includes an address associated with the device, further comprising: determining, by the computing device, that the address associated with the device corresponds to the stored data.
8. An apparatus comprising: one or more processors; and memory storing instructions that, when executed by the one or more processors, cause the apparatus to: send, to a network device located at a premises, a request configured to cause the network device to cause a device located at the premises to send, to the network device, an indication that the device is configured to communicate with the network device at the premises, wherein the premises is associated with a user account that is authorized to access at least one service; receive, from the network device, the indication; based on the indication, determine that the device is located at the premises; and based on determining that the device is located at the premises, grant the device access to the at least one service.
9. The apparatus of claim 8, wherein the instructions that cause the apparatus to send the request further cause the apparatus to send, to the network device, a first prompt message that causes the network device to send a second prompt message to the device, the first prompt message comprising at least in part the request for an indication of devices in communication with the network device.
10. The apparatus of claim 8, wherein the indication comprises an address associated with the device.
11. The apparatus of claim 8, wherein the indication indicates that the device is connected to the network device via a local area network.
12. The apparatus of claim 8, wherein the instructions, when executed by the one or more processors, further cause the apparatus to send, based on the granting the device access to the at least one service, to the network device, content associated with the at least one service to cause the network device to forward the content associated with the at least one service to the device.
13. The apparatus of claim 8, wherein the instructions, when executed by the one or more

processors, further cause the apparatus to store, based on the granting the device access to the at least one service, in a memory associated with the apparatus, data indicating that the device is authorized to access the at least one service.

14. The apparatus of claim 13, wherein the indication includes an address associated with the device, wherein the instructions, when executed by the one or more processors, further cause the apparatus to: determine that the address associated with the device corresponds to the stored data.

15. A non-transitory computer-readable medium storing instructions that, when executed, cause: by a computing device of a service provider that provides at least one service, to a network device located at a premises, a request configured to cause the network device to cause a device located at the premises to send, to the network device, an indication that the device is configured to communicate with the network device at the premises, wherein the premises is associated with a user account that is authorized to access the at least one service; receiving, from the network device, the indication; based on the indication, determining that the device is located at the premises; and based on determining that the device is located at the premises, granting the device access to the at least one service.

16. The non-transitory computer-readable medium of claim 15, wherein the instructions that, when executed, cause the causing sending the request to further cause the device to send, to the network device, a first prompt message that causes the network device to send a second prompt message to the device, the first prompt message comprising at least in part the request for an indication of devices in communication with the network device.

17. The non-transitory computer-readable medium of claim 15, wherein the indication comprises an address associated with the device.

18. The non-transitory computer-readable medium of claim 15, wherein the indication indicates that the device is connected to the network device via a local area network.

19. The non-transitory computer-readable medium of claim 15, wherein the instructions, when executed, further cause sending, based on the granting the device access to the at least one service, to the network device, content associated with the at least one service to cause the network device to forward the content associated with the at least one service to the device.

20. The non-transitory computer-readable medium of claim 15, wherein the instructions, when executed, further cause storing, based on the granting the device access to the at least one service, in a memory associated with the computing device, data indicating that the device is authorized to access the at least one service.

21. The non-transitory computer-readable medium of claim 20, wherein the indication includes an address associated with the device, wherein the instructions, when executed, further cause: determining, by the computing device, that the address associated with the device corresponds to the stored data.

22. A system comprising: a network device located at a premises and configured to: receive, from a device, an indication that the device is configured to communicate with the network device at the premises, and send, to a computing device, the indication; and the computing device configured to: send, to the network device located at the premises, a request configured to cause the network device to cause a device located at the premises to send, to the network device, the indication, wherein the premises is associated with a user account that is authorized to access at least one service, receiving, from the network device, the indication, based on the indication, determine that the device is located at the premises, and based on determining that the device is located at the premises, grant the device access to the at least one service.

23. The system of claim 22, wherein the computing device configured to send the request is further configured to send, to the network device, a first prompt message that causes the network device to send a second prompt message to the device, the first prompt message comprising at least in part the request for an indication of devices in communication with the network device.

24. The system of claim 22, wherein the indication comprises an address associated with the

device.

25. The system of claim 22, wherein the indication indicates that the device is connected to the network device via a local area network.

26. The system of claim 22, wherein the computing device is further configured to send, based on the granting the device access to the at least one service, to the network device, content associated with the at least one service to cause the network device to forward the content associated with the at least one service to the device.

27. The system of claim 22, wherein the computing device is further configured to store, based on the granting the device access to the at least one service, in a memory associated with the computing device, data indicating that the device is authorized to access the at least one service.

28. The system of claim 27, wherein the indication includes an address associated with the device, wherein the computing device is further configured to: determine that the address associated with the device corresponds to the stored data.
